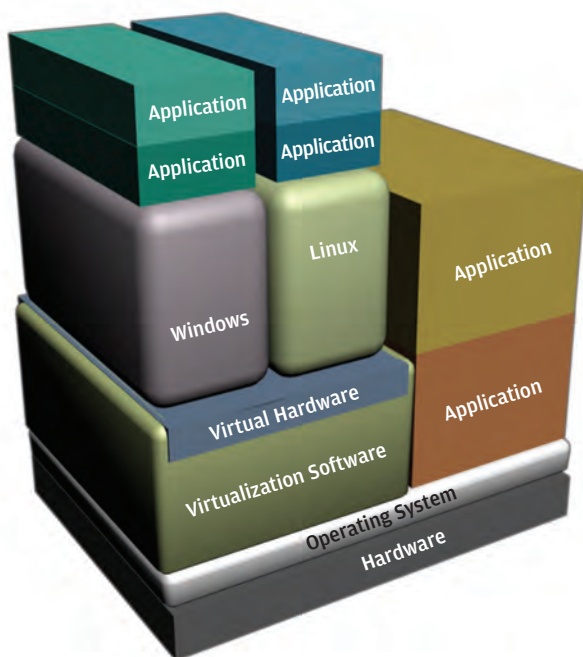


Server utilisation would commonly be at around 12% in most organisations

In a recent report, McKinsey is more gung-ho about virtualization than cloud computing

The virtualization concept dates back to the days of Mainframe computing



What are its advantages?

An obvious benefit of consolidating hardware in such a way is cost saving. Consolidation of IT resources improves server utilization and consequently increases efficiency, lowers power consumption, and reduces the need for physical real estate.

Since hardware resources are utilised more efficiently, the compute capacity on each physical server is maximised. Thus the utilisation of resources increases by at least 5 to 15 per cent. It is important to note that savings increase proportionately with the number of servers that are consolidated by a business. If only a handful of servers are deployed as VMs, then savings will be significantly smaller. Some virtualization platforms claim to have a virtual consolidation ratio of 10:1. Consolidating servers results in increased ROI for both existing and future server expenditures.

With virtualization, each server's programs, files, and processes are merged into one file, and this file then placed among other images on one physical server. This means that software can easily be relocated bringing in an important aspect of portability. If upgrading to a better IT infrastructure the VMs can easily be moved without having to go through the lengthy process of setting up systems all over again.

Small businesses sometimes even deploy virtualization as a way to simplify disaster recovery. Conventionally, disaster recovery can be quite a harrowing ordeal to any IT admin personnel. Issues such as finding the right backup disks, offloading them on to new hardware, and encountering all sorts of problems such as not having a record of software licenses and not being able to find the original CDs for their operating systems and other applications are commonplace. Then comes the cumbersome process of re-installing and re-configuring everything and making it work on the new hardware. Virtualization on the other hand makes it easy to capture everything onto a single image file which makes recovery easier.

Another consequence of this attribute is that it becomes easy to quickly respond to growing business needs. Since virtual servers can be deployed on blade servers, which themselves are modular, it becomes easy to scale up the IT infrastructure.

Testing is another area where virtualization has its uses. In a virtual environment, virtual test servers can be quickly setup, tested and broken down or reconfigured. Some organizations are still running applications on legacy operating systems, which with the help of virtualized servers can now be run on the latest

hardware. Security is also somewhat heightened. System admins go to great lengths to isolate each of the machines in a network by a complex system of firewalls and other measures. Now virtualized servers can reside on the same physical server and yet remain in their own encapsulated environment.

Your roadmap to virtualization

Now that you know the benefits of virtualization it is important to know a few points that will act as guidelines for your enterprise to enter the virtualized world.

Assess your needs: the first step in a virtualization deployment programme should be to assess your needs. Make a business case for virtualization and determine whether the technology will indeed provide a return on investment. Businesses should look at how their current servers are being utilised and answer the following questions

- Are common applications running on several different servers?
- Are there enough servers that could and should be consolidated?
- Is the business's server requirement likely to increase?

Licensing and tech support: it is important to check if current software licenses can be migrated to the virtualized servers. Also, some application developers may be unwilling to provide technical support to their applications when they're moved to a virtualized environment.

Virtualization deployment plan:

an SMB must understand the importance of a sound plan. Timelines, budgets and an implementation programme should be chalked out. Backing up data is very important. Also, the plan should be technically sound. For instance, it is important to assess the hardware requirements of each virtual machine. i.e. how much hardware is needed for it to operate smoothly. The number of VMs should not be allowed to exceed what the hardware is able to support. A thorough implementation plan will help the enterprise minimise glitches. ■

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Application Virtualization

Application virtualization is an interesting variant of virtualization. Unlike server virtualization, where an isolation layer is built around each OS, here, an isolation layer is built around each application. So, in effect, the application is wrapped

or packaged by a layer that does not allow it to freely interact with the underlying OS. This way the application is fooled into taking registry entries and other information from the virtual layer around it and hence the application cannot damage the OS. Virtualized applications can therefore run alongside each other perfectly well. In fact, different versions can also run alongside each other without causing any conflicts. Hence, every application is a nice little bundle that can be very easily distributed. In effect the virtualized program can be streamed or deployed from one central server to clients. Error resolution becomes simpler and so does updating the package. In the context of large workplaces where deployment is a hassle, this makes good sense. Altiris SVS (Software Virtualization Solution), Microsoft Application Virtualization (MAV) and XenApp are some of the AV options available. The way in which each of these solutions creates an abstraction layer differs otherwise they're quite similar in function.

